Evaluation runs: lb (a4-balancer)

generated 2026-11-08 from brain-data 614b86250 2026-11-08 11:41:25 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): hash: expected conforms; hash-drop: expected violation; drop: expected violation (a4a is not a README row); hash: expected conforms; hash-drop: expected violation (a4b is not a README row); drop: expected violation; rr: expected violation; rr: expected conforms (vacuous: a client's requests cannot land on two servers); rr: expected conforms; pair-swap-hash: expected conforms (control; m=1 included. Defined for k=2 only: the balancer reads two requests a round); pair-swap-hash: expected conforms (control; m=1 included. Defined for k=2 only: the balancer reads two requests a round); pair-swap-rr: expected conforms (control; m=1 included. Defined for k=2 only: the balancer reads two requests a round); pair-swap-rr: expected violation (as rr; not a README row. Defined for k=2 only); pair-swap-rr: expected conforms (as rr; not a README row. Defined for k=2 only). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{ref}, G_{cand} = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{ref}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕒 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
1,1,1	1	1	conforms	0.0	4	1 h, 250 GB	1	1	brain03, 2026-11-03
2,1,2	8	2	conforms	0.0	4	1 h, 250 GB	2	9	brain03, 2026-11-03
2,2,2	24	6	conforms	0.0	4	1 h, 250 GB	6	75	brain03, 2026-11-03
3,3,3	45360		🕒 (1 h)			1 h, 250 GB			brain03, 2026-11-03
4,4,4			🕒 (1 h)			1 h, 250 GB			brain03, 2026-11-03

stateless: pair **pair-swap-hash-a4a** against **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
1,1,1	1	1	violation*	0.0	4	1 h, 250 GB	1	1	brain03, 2026-11-03
2,1,2	8	2	conforms	0.0	4	1 h, 250 GB	1	2	brain03, 2026-11-03
2,2,2	24	4	conforms	0.0	4	1 h, 250 GB	1	3	brain03, 2026-11-03
3,3,3	45360	216	violation*	1528.6	12	1 h, 250 GB	216	9797760	brain03, 2026-11-03
4,4,4			🕒 (1 h)			1 h, 250 GB			brain03, 2026-11-03

stateless: pair **pair-swap-rr-a4a** against **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
1,1,1	1	1	violation*	0.0	4	1 h, 250 GB	1	1	brain03, 2026-11-03
2,1,2	8	2	conforms	0.0	4	1 h, 250 GB	2	5	brain03, 2026-11-03
2,2,2	24	4	violation	0.0	4	1 h, 250 GB	4	58	brain03, 2026-11-03
3,3,3	45360	216	violation*	1495.9	13	1 h, 250 GB	216	9797760	brain03, 2026-11-03
4,4,4			🕒 (1 h)			1 h, 250 GB			brain03, 2026-11-03

stateless: pair **rr-a4a** against **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
1,1,1	1	1	conforms	0.0	4	1 h, 250 GB	1	1	brain03, 2026-11-03
2,1,2	8	2	conforms	0.0	2	1 h, 250 GB	2	8	brain03, 2026-11-03
2,2,2	24	6	violation	0.0	2	1 h, 250 GB	6	114	brain03, 2026-11-03
3,3,3	45360		(1 h)			1 h, 250 GB			brain03, 2026-11-03

stateless: pair **drop-a4a** against **hash-a4a**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
1,1,1	1	1	1	violation	0.0	4	1 h, 250 GB	1	1	brain03, 2026-11-03
2,1,2	8	2	2	violation	0.0	4	1 h, 250 GB	2	16	brain03, 2026-11-03
2,2,2	24	6	6	violation	0.0	4	1 h, 250 GB	6	144	brain03, 2026-11-03
3,3,3	45360			(1 h)			1 h, 250 GB			brain03, 2026-11-03

stateless: pair **hash-drop-a4a** against **hash-a4a**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
1,1,1	1	1	1	violation	0.0	4	1 h, 250 GB	1	1	brain03, 2026-11-03
2,1,2	8	2	2	violation	0.0	4	1 h, 250 GB	2	16	brain03, 2026-11-03
2,2,2	24	6	6	violation	0.0	4	1 h, 250 GB	6	144	brain03, 2026-11-03
3,3,3	45360			(1 h)			1 h, 250 GB			brain03, 2026-11-03

stateless: pair **rr-a4a**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
1,2,2	2	1	1	violation	0.0	2	1 h, 250 GB	1	2	brain03, 2026-11-03

Not run (stateless). planned but never scheduled (grid points beyond the ladder): hash-drop a4a 1 1 2 fifo, pair-swap-hash a4a 1 1 2 fifo, drop a4a 1 1 2 fifo, rr a4a 1 1 2 fifo, pair-swap-rr a4a 1 1 2 fifo, hash a4a 1 1 2 fifo, drop a4a 1 1 3 fifo, rr a4a 1 1 3 fifo, pair-swap-rr a4a 1 1 3 fifo, pair-swap-hash a4a 1 1 3 fifo, hash a4a 1 1 3 fifo, hash-drop a4a 1 1 3 fifo, pair-swap-hash a4a 1 1 4 fifo, rr a4a 1 1 4 fifo, hash a4a 1 1 4 fifo, pair-swap-rr a4a 1 1 4 fifo, hash-drop a4a 1 1 4 fifo, drop a4a 1 1 4 fifo, hash a4a 1 2 1 fifo, rr a4a 1 2 1 fifo, pair-swap-hash a4a 1 2 1 fifo, hash-drop a4a 1 2 1 fifo, drop a4a 1 2 1 fifo, pair-swap-rr a4a 1 2 1 fifo, … (358 instances in all).

skipped: a smaller size had reached its limit: hash-drop a4a 4 4 4 fifo, rr a4a 4 4 4 fifo, drop a4a 4 4 4 fifo, pair-swap-rr a4a 5 5 5 fifo, hash a4a 5 5 5 fifo, drop a4a 5 5 5 fifo, pair-swap-hash a4a 5 5 5 fifo, hash-drop a4a 5 5 5 fifo, rr a4a 5 5 5 fifo, rr a4a 6 6 6 fifo, drop a4a 6 6 6 fifo, hash a4a 6 6 6 fifo, hash-drop a4a 6 6 6 fifo, pair-swap-hash a4a 6 6 6 fifo, pair-swap-rr a4a 6 6 6 fifo, pair-swap-hash a4a 7 7 7 fifo, pair-swap-rr a4a 7 7 7 fifo, drop a4a 7 7 7 fifo, hash a4a 7 7 7 fifo, hash-drop a4a 7 7 7 fifo, rr a4a 7 7 7 fifo, pair-swap-hash a4a 8 8 8 fifo, hash-drop a4a 8 8 8 fifo, pair-swap-rr a4a 8 8 8 fifo, … (27 instances in all).

stateful

stateful: pair **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
1,1,1	1	1	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2,1,2	8	2	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2,2,2	24	6	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
3,3,3	45360	1680	conforms	4.6	2	1 h, 250 GB	brain02, 2026-11-03
4,4,4			(1 h)			1 h, 250 GB	brain02, 2026-11-03

stateful: pair **pair-swap-hash-a4a** against **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
1,1,1	1	1	violation*	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2,1,2	8	2	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2,2,2	24	4	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
3,3,3	45360	216	violation*	4.7	2	1 h, 250 GB	brain02, 2026-11-03
4,4,4			(1 h)			1 h, 250 GB	brain02, 2026-11-03

stateful: pair **pair-swap-rr-a4a** against **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
1,1,1	1	1	violation*	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2,1,2	8	2	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2,2,2	24	4	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
3,3,3	45360	216	violation*	5.8	4	1 h, 250 GB	brain02, 2026-11-03
4,4,4			(1 h)			1 h, 250 GB	brain02, 2026-11-03

stateful: pair **rr-a4a** against **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
1,1,1	1	1	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2,1,2	8	2	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2,2,2	24	6	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-03
3,3,3	45360	1680	violation	5.1	16	1 h, 250 GB	brain02, 2026-11-03
4,4,4			(1 h)			1 h, 250 GB	brain02, 2026-11-03

stateful: pair **drop-a4a** against **hash-a4a**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
1,1,1	1	1	1	violation	2.1	2	1 h, 250 GB	brain02, 2026-11-03
2,1,2	8	2	2	violation	2.1	2	1 h, 250 GB	brain02, 2026-11-03
2,2,2	24	6	6	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
3,3,3	45360	1680	1680	violation	5.0	18	1 h, 250 GB	brain02, 2026-11-03
4,4,4				(1 h)			1 h, 250 GB	brain02, 2026-11-03

stateful: pair **hash-drop-a4a** against **hash-a4a**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
1,1,1	1	1	1	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2,1,2	8	2	2	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2,2,2	24	6	6	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
3,3,3	45360	1680	1680	violation	5.3	12	1 h, 250 GB	brain02, 2026-11-03
4,4,4				(1 h)			1 h, 250 GB	brain02, 2026-11-03

stateful: pair **rr-a4a**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
1,2,2	2	1	1	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03

Not run (stateful). planned but never scheduled (grid points beyond the ladder): hash-drop a4a 1 1 2 fifo, pair-swap-hash a4a 1 1 2 fifo, drop a4a 1 1 2 fifo, rr a4a 1 1 2 fifo, pair-swap-rr a4a 1 1 2 fifo, hash a4a 1 1 2 fifo, drop a4a 1 1 3 fifo, rr a4a 1 1 3 fifo, pair-swap-rr a4a 1 1 3 fifo, pair-swap-hash a4a 1 1 3 fifo, hash a4a 1 1 3 fifo, hash-drop a4a 1 1 3 fifo, pair-swap-hash a4a 1 1 4 fifo, rr a4a 1 1 4 fifo, hash a4a 1 1 4 fifo, pair-swap-rr a4a 1 1 4 fifo, hash-drop a4a 1 1 4 fifo, drop a4a 1 1 4 fifo, hash a4a 1 2 1 fifo, rr a4a 1 2 1 fifo, pair-swap-hash a4a 1 2 1 fifo, hash-drop a4a 1 2 1 fifo, drop a4a 1 2 1 fifo, pair-swap-rr a4a 1 2 1 fifo, … (362 instances in all).

skipped: a smaller size had reached its limit: pair-swap-rr a4a 5 5 5 fifo, hash a4a 5 5 5 fifo, drop a4a 5 5 5 fifo, pair-swap-hash a4a 5 5 5 fifo, hash-drop a4a 5 5 5 fifo, rr a4a 5 5 5 fifo, rr a4a 6 6 6 fifo, drop a4a 6 6 6 fifo, hash a4a 6 6 6 fifo, hash-drop

a4a 6 6 6 fifo, pair-swap-hash a4a 6 6 6 fifo, pair-swap-rr a4a 6 6 6 fifo, pair-swap-hash a4a 7 7 7 fifo, pair-swap-rr a4a 7 7 7 fifo, drop a4a 7 7 7 fifo, hash a4a 7 7 7 fifo, hash-drop a4a 7 7 7 fifo, rr a4a 7 7 7 fifo, pair-swap-hash a4a 8 8 8 fifo, hash-drop a4a 8 8 8 fifo, pair-swap-rr a4a 8 8 8 fifo, drop a4a 8 8 8 fifo, rr a4a 8 8 8 fifo, hash a4a 8 8 8 fifo.

offline

offline: pair **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	conforms	13.1	8	10 min, 16 GB	0.1	13.1	0.0	in-memory	brain05, 2026-11-02
2,1,2	8	2	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2,2,2	24	6	conforms	0.2	8	1 h, 250 GB	0.1	0.2	0.0	merge	brain05, 2026-11-03
3,3,3	45360	1680	conforms	7.7	58	4 h, 250 GB	6.2	0.4	1.6	merge	brain05, 2026-11-03

offline: pair **hash-a4a** against **hash-a4b**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	conforms	13.1	8	10 min, 16 GB	0.2	13.1	0.0	in-memory	brain05, 2026-11-02
2,1,2	8	2	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2,2,2	96	6	conforms	0.2	8	1 h, 250 GB	0.1	0.2	0.0	merge	brain05, 2026-11-03
3,3,3	33067440	1680	conforms	1763.1	40085	1 h, 250 GB	784.1	0.3	979.0	merge	brain05, 2026-11-03

offline: pair **pair-swap-hash-a4a** against **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	violation*	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
2,1,2	8	2	conforms	0.4	8	10 min, 16 GB	0.1	0.4	0.0	in-memory	brain05, 2026-11-02
2,2,2	24	4	conforms	0.2	8	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain05, 2026-11-02
3,3,3	45360	216	violation*	7.7	58	4 h, 250 GB	6.2	0.1	1.5	merge	brain05, 2026-11-03

offline: pair **pair-swap-hash-a4a** against **hash-a4b**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	violation*	0.2	8	10 min, 16 GB	0.2	0.0	0.0	in-memory	brain05, 2026-11-02
2,1,2	8	2	conforms	0.4	8	10 min, 16 GB	0.1	0.4	0.0	in-memory	brain05, 2026-11-02
2,2,2	96	4	conforms	0.2	8	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain05, 2026-11-02
3,3,3	33067440	216	(1 h)			1 h, 250 GB	784.1	0.1		in-memory	brain05, 2026-11-03

offline: pair **pair-swap-rr-a4a** against **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	violation*	0.4	8	10 min, 16 GB	0.1	0.4	0.0	in-memory	brain05, 2026-11-02
2,1,2	8	2	conforms	0.3	8	10 min, 16 GB	0.1	0.3	0.0	in-memory	brain05, 2026-11-02
2,2,2	24	4	violation	0.2	8	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain05, 2026-11-02
3,3,3	45360	216	violation*	8.0	58	4 h, 250 GB	6.2	0.1	1.8	merge	brain05, 2026-11-03

offline: pair **pair-swap-rr-a4a** against **hash-a4b**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	violation*	0.4	8	10 min, 16 GB	0.2	0.4	0.0	in-memory	brain05, 2026-11-02
2,1,2	8	2	conforms	0.3	8	10 min, 16 GB	0.1	0.3	0.0	in-memory	brain05, 2026-11-02
2,2,2	96	4	conforms	0.2	8	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain05, 2026-11-02
3,3,3	33067440	216	(1 h)			1 h, 250 GB	784.1	0.1		in-memory	brain05, 2026-11-03

offline: pair **rr-a4a** against **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2,1,2	8	2	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2,2,2	24	6	violation	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
3,3,3	45360	1680	violation	7.7	60	4 h, 250 GB	6.2	0.3	1.5	merge	brain05, 2026-11-03

offline: pair **rr-a4a** against **hash-a4b**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	conforms	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
2,1,2	8	2	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2,2,2	96	6	conforms	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
3,3,3	33067440	1680	(1 h)			1 h, 250 GB	784.1	0.3		in-memory	brain05, 2026-11-03

offline: pair **drop-a4a** against **hash-a4a**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	1	violation	0.6	8	10 min, 16 GB	0.1	0.6	0.0	in-memory	brain05, 2026-11-02
2,1,2	8	2	2	violation	0.1	8	1 h, 250 GB	0.1	0.1	0.0	merge	brain05, 2026-11-03
2,2,2	24	6	6	violation	0.1	8	1 h, 250 GB	0.1	0.1	0.0	merge	brain05, 2026-11-03
3,3,3	45360	1680	1680	violation	7.8	60	4 h, 250 GB	6.2	0.3	1.6	merge	brain05, 2026-11-03

offline: pair **drop-a4a** against **hash-a4b**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	1	violation	0.6	8	10 min, 16 GB	0.2	0.6	0.0	in-memory	brain05, 2026-11-02
2,1,2	8	2	2	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2,2,2	96	6	6	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3,3,3	33067440	1680	1680	violation	1751.5	40085	1 h, 250 GB	784.1	0.3	967.5	merge	brain05, 2026-11-03

offline: pair **hash-drop-a4a** against **hash-a4a**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	1	violation	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
2,1,2	8	2	2	violation	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
2,2,2	24	6	6	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3,3,3	45360	1680	1680	violation	7.7	60	4 h, 250 GB	6.2	0.3	1.5	merge	brain05, 2026-11-03

offline: pair **hash-drop-a4a** against **hash-a4b**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	1	violation	0.2	8	10 min, 16 GB	0.2	0.0	0.0	in-memory	brain05, 2026-11-02
2,1,2	8	2	2	violation	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
2,2,2	96	6	6	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3,3,3	33067440	1680		(1 h)			1 h, 250 GB	784.1	0.3		in-memory	brain05, 2026-11-03

offline: pair **rr-a4a**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,2,2	2	1	1	violation	0.3	8	10 min, 16 GB	0.1	0.3	0.0	in-memory	brain05, 2026-11-02

Not run (offline). planned but never scheduled (grid points beyond the ladder): hash-drop a4a 1 1 2 fifo, pair-swap-hash a4a 1 1 2 fifo, drop a4a 1 1 2 fifo, rr a4a 1 1 2 fifo, pair-swap-rr a4a 1 1 2 fifo, hash a4a 1 1 2 fifo, drop a4a 1 1 3 fifo, rr a4a 1 1 3 fifo, pair-swap-rr a4a 1 1 3 fifo, pair-swap-hash a4a 1 1 3 fifo, hash a4a 1 1 3 fifo, hash-drop a4a 1 1 3 fifo, pair-swap-hash a4a 1 1 4 fifo, rr a4a 1 1 4 fifo, hash a4a 1 1 4 fifo, pair-swap-rr a4a 1 1 4 fifo, hash-drop a4a 1 1 4 fifo, drop a4a 1 1 4 fifo, hash a4a 1 2 1 fifo, rr a4a 1 2 1 fifo, pair-swap-hash a4a 1 2 1 fifo, hash-drop a4a 1 2 1 fifo, drop a4a 1 2 1 fifo, pair-swap-rr a4a 1 2 1 fifo, ... (373 instances in all).

refused (an input missing at run time): pair-swap-hash a4a 4 4 4 fifo, pair-swap-rr a4a 4 4 4 fifo, hash a4a 4 4 4 fifo, pair-swap-rr a4a 5 5 5 fifo, pair-swap-hash a4a 5 5 5 fifo.

skipped: a smaller size had reached its limit: hash-drop a4a 4 4 4 fifo, rr a4a 4 4 4 fifo, drop a4a 4 4 4 fifo, hash a4a 5 5 5 fifo, drop a4a 5 5 5 fifo, hash-drop a4a 5 5 5 fifo, rr a4a 5 5 5 fifo, rr a4a 6 6 6 fifo, drop a4a 6 6 6 fifo, hash a4a 6 6 6 fifo, hash-drop a4a 6 6 6 fifo, pair-swap-hash a4a 6 6 6 fifo, pair-swap-rr a4a 6 6 6 fifo, pair-swap-hash a4a 7 7 7 fifo, pair-swap-rr a4a 7 7 7 fifo, drop a4a 7 7 7 fifo, hash a4a 7 7 7 fifo, hash-drop a4a 7 7 7 fifo, rr a4a 7 7 7 fifo, pair-swap-hash a4a 8 8 8 fifo, hash-drop a4a 8 8 8 fifo, pair-swap-rr a4a 8 8 8 fifo, drop a4a 8 8 8 fifo, rr a4a 8 8 8 fifo, ... (25 instances in all).

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	conforms	0.6	178	10 min, 16 GB	0.1	0.6	0.0	in-memory	brain01, 2026-11-01
2,1,2	8	2	conforms	3.1	176	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-01
2,2,2	24	6	conforms	3.1	166	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-01

cross offline, Java implementation against the Rust specification: pair **hash-a4a** against **hash-a4b**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	conforms	0.6	178	10 min, 16 GB	0.2	0.6	0.0	in-memory	brain01, 2026-11-01
2,1,2	8	2	conforms	3.1	176	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-01
2,2,2	96	6	conforms	3.1	166	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-01

cross offline, Java implementation against the Rust specification: pair **pair-swap-hash-a4a** against **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	violation*	0.6	155	10 min, 16 GB	0.1	0.6	0.0	in-memory	brain01, 2026-11-01
2,1,2	8	2	conforms	2.9	166	10 min, 16 GB	0.1	2.9	0.0	in-memory	brain01, 2026-11-01
2,2,2	24	4	conforms	3.1	184	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-01

cross offline, Java implementation against the Rust specification: pair **pair-swap-hash-a4a** against **hash-a4b**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	violation*	0.6	155	10 min, 16 GB	0.2	0.6	0.0	in-memory	brain01, 2026-11-01
2,1,2	8	2	conforms	2.9	166	10 min, 16 GB	0.1	2.9	0.0	in-memory	brain01, 2026-11-01
2,2,2	96	4	conforms	3.1	184	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-01

cross offline, Java implementation against the Rust specification: pair **pair-swap-rr-a4a** against **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	violation*	0.6	156	10 min, 16 GB	0.1	0.6	0.0	in-memory	brain01, 2026-11-01
2,1,2	8	2	conforms	2.9	162	10 min, 16 GB	0.1	2.9	0.0	in-memory	brain01, 2026-11-01
2,2,2	24	4	violation	3.0	166	10 min, 16 GB	0.1	3.0	0.0	in-memory	brain01, 2026-11-01

cross offline, Java implementation against the Rust specification: pair pair-swap-rr-a4a against hash-a4b, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	violation*	0.6	156	10 min, 16 GB	0.2	0.6	0.0	in-memory	brain01, 2026-11-01
2,1,2	8	2	conforms	2.9	162	10 min, 16 GB	0.1	2.9	0.0	in-memory	brain01, 2026-11-01
2,2,2	96	4	conforms	3.0	166	10 min, 16 GB	0.1	3.0	0.0	in-memory	brain01, 2026-11-01

cross offline, Java implementation against the Rust specification: pair rr-a4a against hash-a4a, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	conforms	0.6	154	10 min, 16 GB	0.1	0.6	0.0	in-memory	brain01, 2026-11-01
2,1,2	8	2	conforms	3.1	162	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-01
2,2,2	24	6	violation	3.1	183	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-01

cross offline, Java implementation against the Rust specification: pair rr-a4a against hash-a4b, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	conforms	0.6	154	10 min, 16 GB	0.2	0.6	0.0	in-memory	brain01, 2026-11-01
2,1,2	8	2	conforms	3.1	162	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-01
2,2,2	96	6	conforms	3.1	183	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-01

cross offline, Java implementation against the Rust specification: pair drop-a4a against hash-a4a, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	1	violation	0.6	154	10 min, 16 GB	0.1	0.6	0.0	in-memory	brain01, 2026-11-01
2,1,2	8	2	2	violation	2.9	180	10 min, 16 GB	0.1	2.9	0.0	in-memory	brain01, 2026-11-01
2,2,2	24	6	6	violation	3.1	165	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-01
2,3,2	80	20	20	violation	5.7	175	48 h, 80 GB	0.1	5.7	0.0	merge	pinky16, 2026-11-07
3,2,2	720	90	90	violation	5.4	190	48 h, 80 GB	0.2	5.4	0.0	merge	pinky29, 2026-11-07

cross offline, Java implementation against the Rust specification: pair drop-a4a against hash-a4b, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	1	violation	0.6	154	10 min, 16 GB	0.2	0.6	0.0	in-memory	brain01, 2026-11-01
2,1,2	8	2	2	violation	2.9	180	10 min, 16 GB	0.1	2.9	0.0	in-memory	brain01, 2026-11-01
2,2,2	96	6	6	violation	3.1	165	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-01
3,2,2	5760	90	90	violation	5.5	197	48 h, 80 GB	0.8	5.4	0.1	merge	pinky29, 2026-11-07

cross offline, Java implementation against the Rust specification: pair hash-drop-a4a against hash-a4a, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	1	violation	0.6	163	10 min, 16 GB	0.1	0.6	0.0	in-memory	brain01, 2026-11-01
2,1,2	8	2	2	violation	2.9	164	10 min, 16 GB	0.1	2.9	0.0	in-memory	brain01, 2026-11-01
2,2,2	24	6	6	violation	3.2	162	10 min, 16 GB	0.1	3.2	0.0	in-memory	brain01, 2026-11-01

cross offline, Java implementation against the Rust specification: pair **hash-drop-a4a** against **hash-a4b**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	1	violation	0.6	163	10 min, 16 GB	0.2	0.6	0.0	in-memory	brain01, 2026-11-01
2,1,2	8	2	2	violation	2.9	164	10 min, 16 GB	0.1	2.9	0.0	in-memory	brain01, 2026-11-01
2,2,2	96	6	6	violation	3.2	162	10 min, 16 GB	0.1	3.2	0.0	in-memory	brain01, 2026-11-01

cross offline, Java implementation against the Rust specification: pair **rr-a4a**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,2,2	2	1	1	violation	0.7	158	10 min, 16 GB	0.1	0.7	0.0	in-memory	brain01, 2026-11-02
1,2,3	3	1	1	violation	0.8	162	48 h, 80 GB	0.1	0.8	0.0	merge	pinky27, 2026-11-07
2,2,2	24	6	4	violation	4.0	193	48 h, 200 GB	0.0	4.0	0.0	merge	volta05, 2026-11-07

Not run (cross offline, Java implementation against the Rust specification). planned but never scheduled (grid points beyond the ladder): rr a4a 1 3 2 fifo, hash-drop a4a 2 2 3 fifo, pair-swap-hash a4a 2 2 3 fifo, drop a4a 2 2 3 fifo, pair-swap-rr a4a 2 2 3 fifo, rr a4a 2 2 3 fifo, hash a4a 2 2 3 fifo, rr a4a 2 3 2 fifo, pair-swap-hash a4a 2 3 2 fifo, drop a4a 2 3 2 fifo, hash-drop a4a 2 3 2 fifo, hash a4a 2 3 2 fifo, pair-swap-rr a4a 2 3 2 fifo, pair-swap-rr a4a 3 2 2 fifo, rr a4a 3 2 2 fifo, hash a4a 3 2 2 fifo, hash-drop a4a 3 2 2 fifo, pair-swap-hash a4a 3 2 2 fifo.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	conforms	13.1	154	10 min, 16 GB	0.7	13.1	0.0	in-memory	brain01, 2026-11-01
2,1,2	8	2	conforms	3.1	167	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-01
2,2,2	24	6	conforms	3.3	175	10 min, 16 GB	3.3	0.2	0.0	in-memory	brain01, 2026-11-01
3,2,2	720	90	conforms	7.3	184	48 h, 250 GB	7.3	0.0	0.0	merge	brain13, 2026-11-07

cross offline, Rust implementation against the Java specification: pair **hash-a4a** against **hash-a4b**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	conforms	13.1	160	10 min, 16 GB	0.7	13.1	0.0	in-memory	brain01, 2026-11-01
2,1,2	8	2	conforms	3.1	166	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-01
2,2,2	96	6	conforms	3.6	172	10 min, 16 GB	3.6	0.2	0.0	in-memory	brain01, 2026-11-01
3,2,2	5760	90	conforms	13.6	3220	48 h, 250 GB	13.5	0.0	0.1	merge	brain13, 2026-11-07

cross offline, Rust implementation against the Java specification: pair **pair-swap-hash-a4a** against **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	violation*	0.7	154	10 min, 16 GB	0.7	0.0	0.0	in-memory	brain01, 2026-11-01
2,1,2	8	2	conforms	3.1	167	10 min, 16 GB	3.1	0.4	0.0	in-memory	brain01, 2026-11-01
2,2,2	24	4	conforms	3.3	175	10 min, 16 GB	3.3	0.2	0.0	in-memory	brain01, 2026-11-01
3,2,2	720	36	violation*	7.3	184	48 h, 250 GB	7.3	0.0	0.0	merge	brain13, 2026-11-07

cross offline, Rust implementation against the Java specification: pair **pair-swap-hash-a4a** against **hash-a4b**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	violation*	0.7	160	10 min, 16 GB	0.7	0.0	0.0	in-memory	brain01, 2026-11-01
2,1,2	8	2	conforms	3.1	166	10 min, 16 GB	3.1	0.4	0.0	in-memory	brain01, 2026-11-01
2,2,2	96	4	conforms	3.6	172	10 min, 16 GB	3.6	0.2	0.0	in-memory	brain01, 2026-11-01
3,2,2	5760	36	violation*	13.6	3220	48 h, 250 GB	13.5	0.0	0.1	merge	brain13, 2026-11-07

cross offline, Rust implementation against the Java specification: pair pair-swap-rr-a4a against hash-a4a, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	violation*	0.7	154	10 min, 16 GB	0.7	0.4	0.0	in-memory	brain01, 2026-11-01
2,1,2	8	2	conforms	3.1	168	10 min, 16 GB	3.1	0.3	0.0	in-memory	brain01, 2026-11-01
2,2,2	24	4	violation	3.3	175	10 min, 16 GB	3.3	0.2	0.0	in-memory	brain01, 2026-11-01
3,2,2	720	36	violation*	7.3	184	48 h, 250 GB	7.3	0.0	0.0	merge	brain13, 2026-11-07

cross offline, Rust implementation against the Java specification: pair pair-swap-rr-a4a against hash-a4b, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	violation*	0.7	160	10 min, 16 GB	0.7	0.4	0.0	in-memory	brain01, 2026-11-01
2,1,2	8	2	conforms	3.1	166	10 min, 16 GB	3.1	0.3	0.0	in-memory	brain01, 2026-11-01
2,2,2	96	4	conforms	3.6	172	10 min, 16 GB	3.6	0.2	0.0	in-memory	brain01, 2026-11-01
3,2,2	5760	36	violation*	13.6	3220	48 h, 250 GB	13.5	0.0	0.1	merge	brain13, 2026-11-07

cross offline, Rust implementation against the Java specification: pair rr-a4a against hash-a4a, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	conforms	0.7	154	10 min, 16 GB	0.7	0.1	0.0	in-memory	brain01, 2026-11-01
2,1,2	8	2	conforms	3.1	167	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-01
2,2,2	24	6	violation	3.3	175	10 min, 16 GB	3.3	0.0	0.0	in-memory	brain01, 2026-11-01
3,2,2	720	90	violation	7.3	184	48 h, 250 GB	7.3	0.0	0.0	merge	brain13, 2026-11-07

cross offline, Rust implementation against the Java specification: pair rr-a4a against hash-a4b, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	conforms	0.7	160	10 min, 16 GB	0.7	0.1	0.0	in-memory	brain01, 2026-11-01
2,1,2	8	2	conforms	3.1	166	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-01
2,2,2	96	6	conforms	3.6	172	10 min, 16 GB	3.6	0.0	0.0	in-memory	brain01, 2026-11-01
3,2,2	5760	90	conforms	13.6	3220	48 h, 250 GB	13.5	0.0	0.1	merge	brain13, 2026-11-07

cross offline, Rust implementation against the Java specification: pair drop-a4a against hash-a4a, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	1	violation	0.7	154	10 min, 16 GB	0.7	0.6	0.0	in-memory	brain01, 2026-11-01
2,1,2	8	2	2	violation	3.1	167	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-01
2,2,2	24	6	6	violation	3.3	175	10 min, 16 GB	3.3	0.1	0.0	in-memory	brain01, 2026-11-01
3,2,2	720	90	90	violation	7.3	184	48 h, 250 GB	7.3	0.0	0.0	merge	brain13, 2026-11-07

cross offline, Rust implementation against the Java specification: pair **drop-a4a** against **hash-a4b**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	1	violation	0.7	160	10 min, 16 GB	0.7	0.6	0.0	in-memory	brain01, 2026-11-01
2,1,2	8	2	2	violation	3.1	166	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-01
2,2,2	96	6	6	violation	3.6	172	10 min, 16 GB	3.6	0.1	0.0	in-memory	brain01, 2026-11-01
3,2,2	5760	90	90	violation	13.6	3220	48 h, 250 GB	13.5	0.0	0.1	merge	brain13, 2026-11-07

cross offline, Rust implementation against the Java specification: pair **hash-drop-a4a** against **hash-a4a**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	1	violation	0.7	154	10 min, 16 GB	0.7	0.0	0.0	in-memory	brain01, 2026-11-01
2,1,2	8	2	2	violation	3.1	168	10 min, 16 GB	3.1	0.0	0.0	in-memory	brain01, 2026-11-01
2,2,2	24	6	6	violation	3.3	175	10 min, 16 GB	3.3	0.1	0.0	in-memory	brain01, 2026-11-01
3,2,2	720	90	90	violation	7.3	184	48 h, 250 GB	7.3	0.0	0.0	merge	brain13, 2026-11-07

cross offline, Rust implementation against the Java specification: pair **hash-drop-a4a** against **hash-a4b**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	1	violation	0.7	160	10 min, 16 GB	0.7	0.0	0.0	in-memory	brain01, 2026-11-01
2,1,2	8	2	2	violation	3.1	166	10 min, 16 GB	3.1	0.0	0.0	in-memory	brain01, 2026-11-01
2,2,2	96	6	6	violation	3.6	172	10 min, 16 GB	3.6	0.1	0.0	in-memory	brain01, 2026-11-01
3,2,2	5760	90	90	violation	13.6	3220	48 h, 250 GB	13.5	0.0	0.1	merge	brain13, 2026-11-07

cross offline, Rust implementation against the Java specification: pair **rr-a4a**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,2,2	2	1	1	violation	2.9	166	10 min, 16 GB	2.9	0.3	0.0	in-memory	brain01, 2026-11-01

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): rr a4a 1 2 3 fifo, rr a4a 1 3 2 fifo, rr a4a 2 2 2 fifo, hash-drop a4a 2 2 3 fifo, pair-swap-rr a4a 2 2 3 fifo, pair-swap-hash a4a 2 2 3 fifo, drop a4a 2 2 3 fifo, rr a4a 2 2 3 fifo, hash a4a 2 2 3 fifo, rr a4a 2 3 2 fifo, pair-swap-hash a4a 2 3 2 fifo, drop a4a 2 3 2 fifo, hash-drop a4a 2 3 2 fifo, hash a4a 2 3 2 fifo, pair-swap-rr a4a 2 3 2 fifo.

cross online, Java implementation (JMC) against the Rust specification export

cross online, Java implementation (JMC) against the Rust specification export: pair **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	conforms	0.7	146	1 h, 250 GB		brain05, 2026-11-02
2,1,2	8	2	conforms	0.7	150	1 h, 250 GB		brain05, 2026-11-02
2,2,2	24	6	conforms	0.8	152	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **hash-a4a** against **hash-a4b**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	conforms	0.8	146	1 h, 250 GB		brain05, 2026-11-02
2,1,2	8	2	conforms	0.7	152	1 h, 250 GB		brain05, 2026-11-02
2,2,2	96	6	conforms	0.8	156	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair pair-swap-hash-a4a against hash-a4a, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	violation*	0.7	148	1 h, 250 GB	43	brain05, 2026-11-02
2,1,2	8	2	conforms	0.7	148	1 h, 250 GB		brain05, 2026-11-02
2,2,2	24	4	conforms	0.8	152	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair pair-swap-hash-a4a against hash-a4b, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	violation*	0.8	156	1 h, 250 GB	42	brain05, 2026-11-02
2,1,2	8	2	conforms	0.6	150	1 h, 250 GB		brain05, 2026-11-02
2,2,2	96	4	conforms	0.7	156	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair pair-swap-rr-a4a against hash-a4a, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	violation*	0.7	146	1 h, 250 GB	31	brain05, 2026-11-02
2,1,2	8	2	conforms	0.7	148	1 h, 250 GB		brain05, 2026-11-02
2,2,2	24	4	violation	0.8	148	1 h, 250 GB	86	brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair pair-swap-rr-a4a against hash-a4b, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	violation*	0.8	147	1 h, 250 GB	35	brain05, 2026-11-02
2,1,2	8	2	conforms	0.6	148	1 h, 250 GB		brain05, 2026-11-02
2,2,2	96	4	conforms	0.7	154	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair rr-a4a against hash-a4a, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	conforms	0.7	150	1 h, 250 GB		brain05, 2026-11-02
2,1,2	8	2	conforms	0.7	144	1 h, 250 GB		brain05, 2026-11-02
2,2,2	24	6	violation	0.8	148	1 h, 250 GB	83	brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair rr-a4a against hash-a4b, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	conforms	0.8	148	1 h, 250 GB		brain05, 2026-11-02
2,1,2	8	2	conforms	0.6	150	1 h, 250 GB		brain05, 2026-11-02
2,2,2	96	6	conforms	0.8	156	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair drop-a4a against hash-a4a, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	1	violation	0.7	144	1 h, 250 GB	38	brain05, 2026-11-02
2,1,2	8	2	2	violation	0.7	144	1 h, 250 GB	45	brain05, 2026-11-02
2,2,2	24	6	6	violation	0.8	156	1 h, 250 GB	55	brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair drop-a4a against hash-a4b, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	1	violation	0.7	154	1 h, 250 GB	36	brain05, 2026-11-02
2,1,2	8	2	2	violation	0.6	146	1 h, 250 GB	43	brain05, 2026-11-02
2,2,2	96	6	6	violation	0.8	174	1 h, 250 GB	53	brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair hash-drop-a4a against hash-a4a, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	1	violation	0.7	152	1 h, 250 GB	39	brain05, 2026-11-02
2,1,2	8	2	2	violation	0.7	177	1 h, 250 GB	45	brain05, 2026-11-02
2,2,2	24	6	6	violation	0.8	148	1 h, 250 GB	52	brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair hash-drop-a4a against hash-a4b, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	1	violation	0.8	158	1 h, 250 GB	39	brain05, 2026-11-02
2,1,2	8	2	2	violation	0.6	148	1 h, 250 GB	38	brain05, 2026-11-02
2,2,2	96	6	6	violation	0.8	152	1 h, 250 GB	44	brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair rr-a4a, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1,2,2	2	1	1	violation	0.6	147	1 h, 250 GB	43	brain05, 2026-11-02

Not run (cross online, Java implementation (JMC) against the Rust specification export). planned but never scheduled (grid points beyond the ladder): hash a4a 1 1 1 fifo, pair-swap-hash a4a 1 1 1 fifo, hash-drop a4a 1 1 1 fifo, pair-swap-rr a4a 1 1 1 fifo, rr a4a 1 1 1 fifo, drop a4a 1 1 1 fifo, rr a4a 1 2 2 fifo, pair-swap-rr a4a 2 1 2 fifo, hash a4a 2 1 2 fifo, pair-swap-hash a4a 2 1 2 fifo, drop a4a 2 1 2 fifo, rr a4a 2 1 2 fifo, hash-drop a4a 2 1 2 fifo, hash a4a 2 2 2 fifo, hash-drop a4a 2 2 2 fifo, drop a4a 2 2 2 fifo, rr a4a 2 2 2 fifo, pair-swap-hash a4a 2 2 2 fifo, pair-swap-rr a4a 2 2 2 fifo.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair hash-a4a, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	conforms	0.7	150	1 h, 250 GB		brain04, 2026-11-01
2,1,2	8	2	conforms	3.1	163	1 h, 250 GB		brain04, 2026-11-01

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2,2	24	6	conforms	3.4	171	1 h, 250 GB		brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair hash-a4a against hash-a4b, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	conforms	0.7	156	1 h, 250 GB		brain04, 2026-11-01
2,1,2	8	2	conforms	3.1	162	1 h, 250 GB		brain04, 2026-11-01
2,2,2	96	6	conforms	3.6	168	1 h, 250 GB		brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair pair-swap-hash-a4a against hash-a4a, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	violation*	0.7	150	1 h, 250 GB	1	brain04, 2026-11-01
2,1,2	8	2	conforms	3.1	163	1 h, 250 GB		brain04, 2026-11-01
2,2,2	24	4	conforms	3.4	171	1 h, 250 GB		brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair pair-swap-hash-a4a against hash-a4b, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	violation*	0.7	156	1 h, 250 GB	1	brain04, 2026-11-01
2,1,2	8	2	conforms	3.1	162	1 h, 250 GB		brain04, 2026-11-01
2,2,2	96	4	conforms	3.6	168	1 h, 250 GB		brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair pair-swap-rr-a4a against hash-a4a, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	violation*	0.7	150	1 h, 250 GB	1	brain04, 2026-11-01
2,1,2	8	2	conforms	3.1	163	1 h, 250 GB		brain04, 2026-11-01
2,2,2	24	4	violation	3.4	171	1 h, 250 GB	2	brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair pair-swap-rr-a4a against hash-a4b, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	violation*	0.7	156	1 h, 250 GB	1	brain04, 2026-11-01
2,1,2	8	2	conforms	3.1	162	1 h, 250 GB		brain04, 2026-11-01
2,2,2	96	4	conforms	3.6	168	1 h, 250 GB		brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair rr-a4a against hash-a4a, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	conforms	0.7	150	1 h, 250 GB		brain04, 2026-11-01
2,1,2	8	2	conforms	3.1	163	1 h, 250 GB		brain04, 2026-11-01
2,2,2	24	6	violation	3.4	171	1 h, 250 GB	2	brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair rr-a4a against hash-a4b, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	conforms	0.7	156	1 h, 250 GB		brain04, 2026-11-01
2,1,2	8	2	conforms	3.1	162	1 h, 250 GB		brain04, 2026-11-01
2,2,2	96	6	conforms	3.6	168	1 h, 250 GB		brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair drop-a4a against hash-a4a, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	1	violation	0.7	150	1 h, 250 GB	2	brain04, 2026-11-01
2,1,2	8	2	2	violation	3.1	163	1 h, 250 GB	1	brain04, 2026-11-01
2,2,2	24	6	6	violation	3.3	171	1 h, 250 GB	2	brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair drop-a4a against hash-a4b, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	1	violation	0.7	156	1 h, 250 GB	2	brain04, 2026-11-01
2,1,2	8	2	2	violation	3.1	162	1 h, 250 GB	2	brain04, 2026-11-01
2,2,2	96	6	6	violation	3.6	168	1 h, 250 GB	2	brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair hash-drop-a4a against hash-a4a, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	1	violation	0.7	150	1 h, 250 GB	1	brain04, 2026-11-01
2,1,2	8	2	2	violation	3.1	163	1 h, 250 GB	1	brain04, 2026-11-01
2,2,2	24	6	6	violation	3.4	171	1 h, 250 GB	2	brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair hash-drop-a4a against hash-a4b, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	1	violation	0.7	156	1 h, 250 GB	1	brain04, 2026-11-01
2,1,2	8	2	2	violation	3.1	162	1 h, 250 GB	2	brain04, 2026-11-01
2,2,2	96	6	6	violation	3.6	168	1 h, 250 GB	2	brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair rr-a4a, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1,2,2	2	1	1	violation	3.0	162	1 h, 250 GB	1	brain04, 2026-11-01

Not run (cross online, Rust implementation (TraceForge) against the Java specification export). left out by the table rules (ok): hash a4a 1 1 1 fifo, pair-swap-hash a4a 1 1 1 fifo, hash-drop a4a 1 1 1 fifo, pair-swap-rr a4a 1 1 1 fifo, rr a4a 1 1 1 fifo, drop a4a 1 1 1 fifo, rr a4a 1 2 2 fifo, pair-swap-rr a4a 2 1 2 fifo, hash a4a 2 1 2 fifo, pair-swap-hash a4a 2 1 2 fifo, drop a4a 2 1 2 fifo, rr a4a 2 1 2 fifo, hash-drop a4a 2 1 2 fifo, hash a4a 2 2 2 fifo, hash-drop a4a 2 2 2 fifo, drop a4a 2 2 2 fifo, rr a4a 2 2 2 fifo, pair-swap-hash a4a 2 2 2 fifo, pair-swap-rr a4a 2 2 2 fifo.